

CT1: Computer engineering

This repository contains the lectures, exercises and laboratories for computer engineering 1.

Do not fork this repository.

See chapter [Setting up a private repository](#) for instructions on how to get a private copy of the repository.

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Terminology

- Path are relative whenever applicable.
- Values in angle brackets (<>) are usually placeholders for setupspecific parameters.

Setting up a private repository

It is sensible to work with Git and have a private copy of this repository.

However if you fork it on Github, students after you will have trouble finding the right repository searching for `ct`. Also they will have your solutions, though experience shows, they will have access to solutions anyways. (It is usually rather easy to spot.)

Thus you shall not use Githubs `fork` function. Instead follow the steps below.

- 1) Clone the CT1 repository locally. Read `git-introduction.(md|pdf)` if you do not know how to use Git.
- 2) Create an empty and **private** repository with your user as owner.

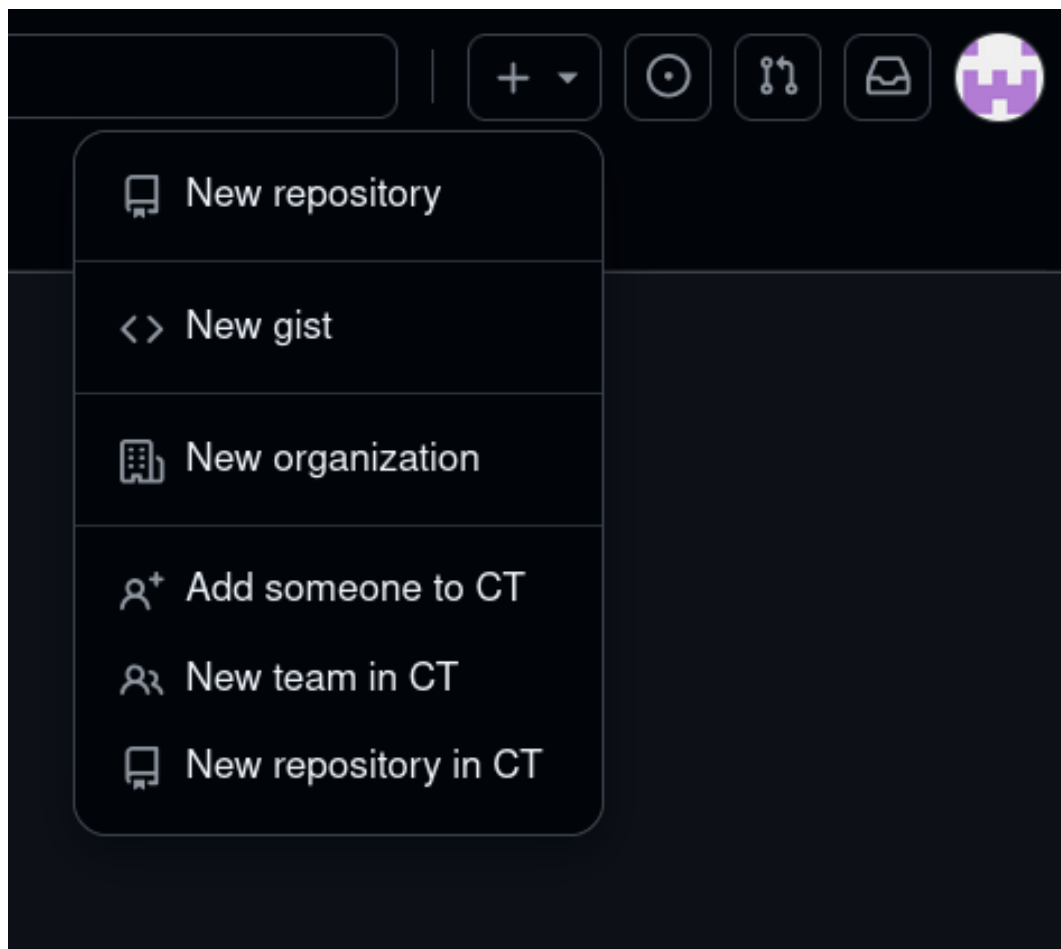


Figure 1: Creating a new repository

Create a new repository

Repositories contain a project's files and version history.
Required fields are marked with an asterisk (*).

1

General

Owner * **Repository name ***

 scbj

 /

ct1

ct1 is available.

Great repository names are short and memorable. How about [musical-octo-funicular](#)?

Description

0 / 350 characters

2

Configuration

Choose visibility *
Choose who can see and commit to this repository

Private

Start with a template
Templates pre-configure your repository with files.

No template

Add README
READMEs can be used as longer descriptions. [About READMEs](#)

Off

Add .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)

No .gitignore

Add license
Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

Figure 2: Settings for new repository

Make sure to select your **user as owner**. Choose a recognisable name for you new repository. The description is optional.

As mentioned before: Set the visibility to **Private**.

- 3) Change the URL of your local copy of the CT1 repository.

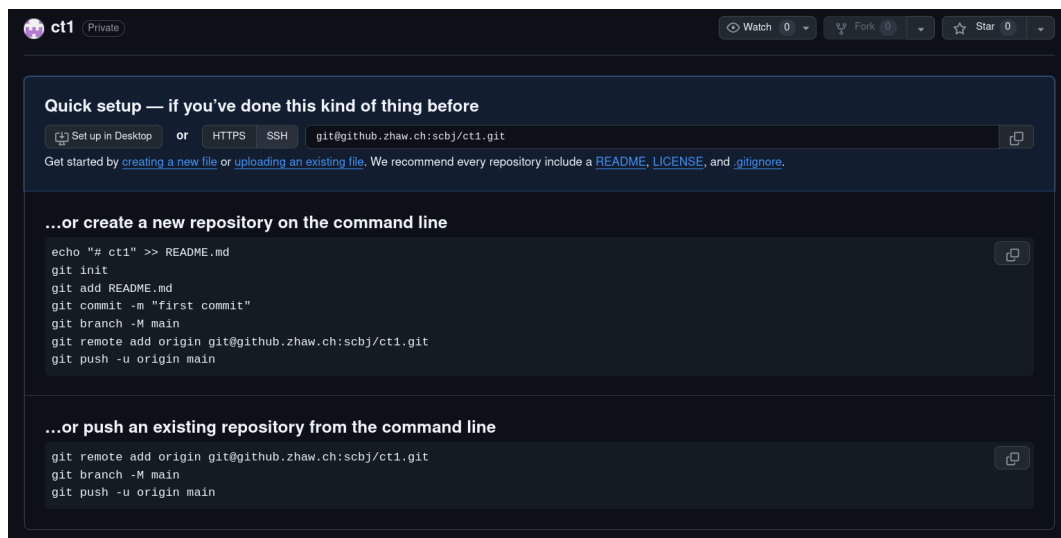


Figure 3: Newly created empty repository

Copy the URL in the 'Quick setup' box. In your local copy of the CT1 repository set the remote URL to your empty repository.

```
1 cd <path/to/ct1/repository>
2 git remote set-url origin <url-to-private-repository>
```

- 4) In order to get updates, set up a second remote for your local copy, called upstream with the URL of the provided repository.

```
1 # synopsis: git remote add <remote-name> <remote-url>
2 git remote add upstream git@github.zhaw.ch:CT/ct1-students
   .git
```

To get updates, you can now use `git pull upstream main`. This will cause conflicts, generally you can choose your own changes on any laboratory you have already solved, and choose the upstream change on all others.

- 5) **Optionally** if you work on the labs with a fellow student, you can open the settings of your private repository and add them as collaborator in order to give them access.

File structure

Below is the file tree of the students repository for CT1. It only shows files which are mentioned in this document.

Entries ending with '/' are directories.

```
1 ct1-students
```

```

2 +-- common/
3 |   +-- .gdbinit
4 |   +-- ct.code-profile
5 |   +-- pdf.mk
6 |   +-- root.mk
7 |   +-- template.tex
8 |   '-- trailing-whitespace.sh
9 +-- exercises/
10 +-- git-introduction/
11 |   +-- Makefile
12 |   +-- readme.md
13 |   '-- readme.pdf
14 +-- laboratories/
15 |   +-- Makefile
16 |   +-- readme.md
17 |   '-- readme.pdf
18 +-- libctboard/
19 +-- theory/
20 +-- Makefile
21 +-- readme.md
22 '-- readme.pdf

```

Every Markdown file should have a `Makefile` with rules to convert from Markdown to PDF (requires `pandoc` and `texlive`). The students releases should contain a PDF version of every Markdown file.

`./readme.(md|pdf)` is the file you are reading. It contains a file tree and instructions on “forking” without forking.

`./laboratories/readme.(md|pdf)` contains instructions on how to setup the required tools for the laboratories, how to work with the makefiles, VS Code or Keil and a short introduction to the GNU debugger.

`./common/.gdbinit` contains commands to be run by the GNU debugger at startup. For example if you want to modify the `ines` layout you could change this file.

`./common/ct.code-profile` is a profile for VS Code users, which comes with the required extensions and settings.

`./common/pdf.mk` is a make include file with rules to convert Markdown to PDF.

`./common/root.mk` is a make include file with rules to build and link binaries from C and assembly sources.

`./common/template.tex` is a LaTeX template for pandoc to customise the PDF output of pandoc.

`./common/trailing-whitespace.sh` is a bash script to detect and remove trailing whitespace and optionally convert line endings to unix style (lf).

`./git-introduction/readme.(md|pdf)` is a short introduction to Git. It is mostly intended for beginners, but also discusses some more advanced topics.